

PROJECTIVE AND INJECTIVE MODULES OVER DEDEKIND DOMAINS

QIXUAN FANG

ABSTRACT. In this term paper, we study the homological properties of several types of modules over a Dedekind domain, including ideals, quotient rings, and finitely generated modules. In this way, we describe several homological features of modules over Dedekind domains.

1. INTRODUCTION

A Dedekind domain is an integrally closed Noetherian domain of Krull dimension at most 1. Thus, under the convention used in this paper, fields are Dedekind domains. The theory grew out of Kummer's work on ideal numbers and Dedekind's subsequent theory of ideals, motivated in part by the failure of unique factorization of elements in rings of algebraic integers. Nowadays, Dedekind domains play an important role in algebraic number theory and algebraic geometry.

Homological algebra developed rapidly in the twentieth century, especially after the 1940s, alongside topology and the theory of homology groups. Modules, particularly modules over specific rings, are among its major objects of study. Homological properties of modules reflect properties of the ring itself, while the structure of the ring conversely influences the modules over it. Although the properties of Dedekind domains have been studied in detail in many classical monographs on commutative algebra and algebraic number theory, in this paper we focus mainly on the module-theoretic point of view.

Ideals and quotient rings are two basic types of modules over a ring, and other modules can often be described in terms of them. We first study the ideals of a Dedekind domain R and obtain several equivalent conditions for an integral domain R to be a Dedekind domain (see [Theorem 1](#)), including:

- (i) every nonzero fractional ideal I of R is invertible;
- (ii) every nonzero fractional ideal is a finitely generated projective R -module;
- (iii) every nonzero fractional ideal I is a finitely generated flat R -module;
- (iv) every nonzero fractional ideal I is finitely generated, and for any prime ideal $\mathfrak{p}$ of R , $I_{\mathfrak{p}}$ is a principal ideal of $R_{\mathfrak{p}}$;
- (v) every nonzero fractional ideal I is finitely generated, and for any maximal ideal $\mathfrak{m}$ of R , $I_{\mathfrak{m}}$ is a principal ideal of $R_{\mathfrak{m}}$.

These equivalent conditions describe the local structure of a Dedekind domain: the localization at each nonzero prime ideal is a discrete valuation ring. Afterwards, we explore the projective property further: every submodule of a projective R -module is projective. This is one way to characterize a hereditary domain.

Having studied ideals, it is natural to continue with quotient rings. First, we give a useful general form of the Chinese Remainder Theorem over a Dedekind domain. We then use it to show that for every nonzero ideal I of R , R/I is a

principal ideal ring, and hence every ideal of R can be generated by at most two elements. Next, we prove a major result of this section: among Noetherian domains that are not fields, R is a Dedekind domain if and only if all its nontrivial quotient rings are quasi-Frobenius rings (QF-rings). This is closely related to injective and self-injective properties. Finally, we explore injectivity further: every quotient of an injective R -module is injective. This is another characteristic property of hereditary rings.

In what follows, we show how ideals and quotient rings describe other modules over R . We continue studying projective and injective modules, and explore some properties of ideals with respect to the functors Hom and $\otimes_R$ and to direct sums. The first major result is that a finitely generated R -module is projective if and only if it is torsion-free. We then discuss the structure and injective envelopes of torsion modules; in particular, indecomposable torsion modules are quasi-injective, and their injective envelopes are described by Matlis' theory. Finally, we discuss the structure theorem for finitely generated modules over a Dedekind domain.

The paper is divided into four sections, studying ideals, quotient rings, and finitely generated modules in [Sections 2](#) to [4](#), respectively. Throughout this paper, R denotes the domain or ring under discussion, and I usually denotes an ideal. We write $S^{-1}R$ for localization at a multiplicative subset S , $\text{Spec}(R)$ for the set of prime ideals of R , and Hom and $\otimes_R$ for the usual module functors.

2. IDEALS OF DEDEKIND DOMAINS AND PROJECTIVE MODULES

We begin by recalling the definition of a Dedekind domain.

Definition 1. A Dedekind domain R is an integrally closed Noetherian domain of Krull dimension at most 1. Thus fields are included as Dedekind domains under our convention.

A direct consequence is the following.

Corollary 1. *Every nonzero prime ideal of a Dedekind domain R is maximal.*

Since localization is a useful way to study rings, we first give a local characterization.

Lemma 1 ([\[Stacks\]](#), Lemma 10.120.17 (Tag 034X)). *Let R be a Noetherian domain. Then R is a Dedekind domain if and only if, for every nonzero maximal ideal $\mathfrak{m}$ of R , $R_{\mathfrak{m}}$ is a discrete valuation ring (DVR).*

This is one of the most useful characterizations of a Dedekind domain. We shall see several forms and applications of it below.

We next recall the unique factorization of ideals in a Dedekind domain.

Lemma 2 ([\[Fen86\]](#), Section 5.3, Theorem 5 and Corollary 2). *An integral domain R is a Dedekind domain if and only if every nonzero proper ideal of R can be uniquely factorized, up to order, as a product of finitely many nonzero prime ideals.*

Since [Lemma 1](#) gives a local characterization of a Dedekind domain, we begin with the corresponding statement for a DVR.

Lemma 3 ([\[AM94\]](#), Proposition 9.7). *Let A be a local domain that is not a field. Then A is a DVR if and only if every nonzero fractional ideal of A is invertible.*

Returning to the global counterpart of [Lemma 3](#), we have the following.

Lemma 4 ([AM94], Theorem 9.8). *Let R be an integral domain. Then R is a Dedekind domain if and only if every nonzero fractional ideal of R is invertible.*

Although this is already a useful characterization, our purpose is to study the homological properties of ideals. We shall need the following lemma.

Lemma 5. *Let I be a nonzero fractional ideal of an integral domain R . Then I is invertible if and only if it is a finitely generated projective R -module.*

Proof. First suppose that I is invertible. Then $II^{-1} = R$, so there are elements $b_1, \dots, b_n \in I$ and $a_1, \dots, a_n \in I^{-1}$ such that

$$1 = \sum_{i=1}^n a_i b_i.$$

In particular, the b_i generate I : for $c \in I$,

$$c = \sum_{i=1}^n (ca_i) b_i,$$

and $ca_i \in R$. Let $F = \bigoplus_{i=1}^n Re_i$ and define $f : F \rightarrow I$ by $f(e_i) = b_i$. Define $g : I \rightarrow F$ by

$$g(c) = \sum_{i=1}^n ca_i e_i.$$

Then $fg = 1_I$. Hence the exact sequence $0 \rightarrow \ker f \rightarrow F \rightarrow I \rightarrow 0$ splits, so I is a finitely generated projective module.

Conversely, suppose that I is finitely generated and projective. By the dual basis lemma, there are $b_1, \dots, b_n \in I$ and $\varphi_1, \dots, \varphi_n \in \text{Hom}_R(I, R)$ such that

$$c = \sum_{i=1}^n \varphi_i(c) b_i \quad (c \in I).$$

Every R -linear map $\varphi : I \rightarrow R$ is multiplication by an element of the quotient field K of R . Indeed, fix $0 \neq b \in I$. For $0 \neq c \in I$, choose nonzero $r, s \in R$ with $sc = rb$. Then

$$s\varphi(c) = r\varphi(b),$$

so $\varphi(c)/c = \varphi(b)/b$. Thus $\varphi(c) = ac$ for a fixed $a \in K$, and since $\varphi(I) \subseteq R$, we have $a \in I^{-1}$. Write $\varphi_i(c) = a_i c$ with $a_i \in I^{-1}$. For any $0 \neq c \in I$,

$$c = c \sum_{i=1}^n a_i b_i,$$

so $1 = \sum_i a_i b_i \in I^{-1}I$. Therefore $I^{-1}I = R$, and I is invertible. $\square$

Our first result is a direct corollary of the preceding lemmas.

Corollary 2. *Let R be an integral domain. Then R is a Dedekind domain if and only if every nonzero fractional ideal is a finitely generated projective R -module.*

We will also use the standard local characterization of finite projective modules.

Lemma 6 ([Stacks], Lemma 10.78.2 (Tag 00NX)). *Let M be a finitely presented module over a commutative ring R . Then M is projective if and only if $M_{\mathfrak{p}}$ is a free $R_{\mathfrak{p}}$ -module for every prime ideal $\mathfrak{p}$ of R ; equivalently, it is enough to check this for every maximal ideal $\mathfrak{m}$ of R .*

Therefore we obtain the following equivalent conditions for fractional ideals.

Lemma 7. *The following are equivalent for a nonzero fractional ideal I over an integral domain R :*

- (i) I is invertible;
- (ii) I is finitely generated projective;
- (iii) I is finitely generated flat;
- (iv) I is finitely generated, and for every prime ideal $\mathfrak{p}$ of R , $I_{\mathfrak{p}}$ is a principal fractional ideal of $R_{\mathfrak{p}}$;
- (v) I is finitely generated, and for every maximal ideal $\mathfrak{m}$ of R , $I_{\mathfrak{m}}$ is a principal fractional ideal of $R_{\mathfrak{m}}$.

Proof. The equivalence of (i) and (ii) is [Lemma 5](#). A finitely generated projective module is flat, so (ii) implies (iii). Suppose (iii) holds. For every maximal ideal $\mathfrak{m}$, the localization $I_{\mathfrak{m}}$ is a finite flat module over the local ring $R_{\mathfrak{m}}$, hence is finite free by [[Stacks](#), Lemma 10.78.5, Tag 00NZ]. Since $I_{\mathfrak{m}}$ is a nonzero fractional ideal, it has rank one, so it is principal. Thus (iii) implies (v). Conversely, if (v) holds, then every $I_{\mathfrak{m}}$ is an invertible fractional ideal of $R_{\mathfrak{m}}$, and [[AM94](#), Proposition 9.6] implies that I is invertible. Thus (v) implies (i). Finally, (ii) implies (iv) by [Lemma 6](#), since a finite projective module is finitely presented and each localization of I has rank one; and (iv) implies (v). This proves the equivalence of all five conditions. $\square$

Combining [Lemmas 4](#) and [7](#) gives the following theorem.

Theorem 1. *The following statements are equivalent for an integral domain R :*

- (i) R is a Dedekind domain;
- (ii) every nonzero fractional ideal I of R is invertible;
- (iii) every nonzero fractional ideal is a finitely generated projective R -module;
- (iv) every nonzero fractional ideal is a finitely generated flat R -module;
- (v) every nonzero fractional ideal I is finitely generated, and for every prime ideal $\mathfrak{p}$ of R , $I_{\mathfrak{p}}$ is a principal fractional ideal of $R_{\mathfrak{p}}$;
- (vi) every nonzero fractional ideal I is finitely generated, and for every maximal ideal $\mathfrak{m}$ of R , $I_{\mathfrak{m}}$ is a principal fractional ideal of $R_{\mathfrak{m}}$.

There is another important homological property of a Dedekind domain.

Theorem 2. *Let R be a Dedekind domain. Then R is a commutative hereditary domain; in particular, every submodule of a projective R -module is projective.*

Proof. By [Theorem 1](#), every ideal of R is projective. A commutative ring is hereditary if and only if every ideal is projective; equivalently, every submodule of a projective module is projective (see [[Fen86](#)], Corollary 4.14 and Theorem 4.19). Therefore R is hereditary. $\square$

3. QUOTIENT RINGS OF DEDEKIND DOMAINS AND INJECTIVE MODULES

We first introduce a useful generalized form of the Chinese Remainder Theorem.

Lemma 8. (*Chinese Remainder Theorem*) *Let $I_1, \dots, I_n$ be nonzero ideals and let $x_1, \dots, x_n$ be elements of a Dedekind domain R . Then the system of congruences*

$$x \equiv x_i \pmod{I_i} \quad (1 \leq i \leq n)$$

has a solution $x \in R$ if and only if

$$x_i \equiv x_j \pmod{I_i + I_j}$$

for every $i \neq j$.

Proof. Consider the sequence

$$R \xrightarrow{\phi} \bigoplus_{i=1}^n R/I_i \xrightarrow{\psi} \bigoplus_{i < j} R/(I_i + I_j),$$

where

$$\phi(x) = (x + I_i)_i, \quad \psi((x_i + I_i)_i) = (x_i - x_j + (I_i + I_j))_{i < j}.$$

It is enough to show that this sequence is exact at the middle term. If R is a field, then every nonzero ideal I_i equals R , and the assertion is immediate. Assume therefore that R is not a field. Localize at a maximal ideal $\mathfrak{m}$. Then $\mathfrak{m} \neq 0$, so $R_{\mathfrak{m}}$ is a DVR by [Lemma 1](#). Write

$$I_i R_{\mathfrak{m}} = (\pi^{e_i})$$

with $e_i \geq 0$, where $e_i = 0$ is allowed if $I_i R_{\mathfrak{m}} = R_{\mathfrak{m}}$. Suppose $(y_i)_i$ is in the kernel of $\psi_{\mathfrak{m}}$, and choose j such that $e_j = \max_i e_i$. Then for every i ,

$$y_i - y_j \in (\pi^{e_i}) + (\pi^{e_j}) = (\pi^{e_i}),$$

so y_j is a common lift of all the y_i . Hence the localized sequence is exact at every maximal ideal. Exactness can be checked locally, so the original sequence is exact. The asserted criterion is precisely the condition that $(x_i + I_i)_i$ lie in $\ker \psi = \operatorname{im} \phi$. $\square$

Corollary 3. *Let R be a Dedekind domain. If $I_1, \dots, I_n$ are pairwise comaximal nonzero ideals of R , then*

$$R / \bigcap_{i=1}^n I_i \cong \prod_{i=1}^n R / I_i.$$

Corollary 4. *Let R be a Dedekind domain. If I is a nonzero ideal of R and*

$$I = \prod_{i=1}^n \mathfrak{p}_i^{m_i},$$

where $\mathfrak{p}_1, \dots, \mathfrak{p}_n$ are distinct nonzero prime ideals, then

$$R/I \cong \prod_{i=1}^n R/\mathfrak{p}_i^{m_i}.$$

We now study the quotient rings more closely. We will use the following proposition.

Proposition 1 ([\[KOS07\]](#), Proposition 6.13). *Let R be a Dedekind domain, let $\mathfrak{q}$ be a nonzero prime ideal of R , and let I be an ideal such that $I \supseteq \mathfrak{q}^m$ for some $m \in \mathbb{N}$. Then $I = \mathfrak{q}^k$ for some $0 \leq k \leq m$. In particular, if $y \in \mathfrak{q} \setminus \mathfrak{q}^2$, then*

$$\mathfrak{q}^t = y^t R + \mathfrak{q}^n$$

for every $n \in \mathbb{N}$ and $1 \leq t \leq n$.

Lemma 9. *Let I be a nonzero ideal of a Dedekind domain R . Then R/I is a principal ideal ring.*

Proof. If $I = R$, then R/I is the zero ring, which is a principal ideal ring. Assume $I \subsetneq R$. Write

$$I = \prod_{i=1}^t \mathfrak{p}_i^{k_i}.$$

By Corollary 4,

$$R/I \cong \prod_{i=1}^t R/\mathfrak{p}_i^{k_i}.$$

An ideal of $R/\mathfrak{p}_i^{k_i}$ has the form $J/\mathfrak{p}_i^{k_i}$ for an ideal J satisfying $\mathfrak{p}_i^{k_i} \subseteq J \subseteq R$. By Proposition 1, $J = \mathfrak{p}_i^s$ for some $0 \leq s \leq k_i$. Choose $y_i \in \mathfrak{p}_i \setminus \mathfrak{p}_i^2$. Proposition 1 gives

$$\mathfrak{p}_i^s = y_i^s R + \mathfrak{p}_i^{k_i},$$

so $J/\mathfrak{p}_i^{k_i}$ is principal. Thus every factor $R/\mathfrak{p}_i^{k_i}$ is a principal ideal ring, and so is their finite direct product. $\square$

Using Lemma 9, we can also see how an ideal is generated from another perspective.

Corollary 5. *Every ideal I in a Dedekind domain R can be generated by at most two elements.*

Proof. If $I = 0$, there is nothing to prove. Otherwise choose $0 \neq a_1 \in I$ and put $J = a_1 R$. By Lemma 9, R/J is a principal ideal ring, so the ideal I/J of R/J is generated by some $a_2 + J$ with $a_2 \in I$. Hence

$$I/J = (a_2 R + J)/J,$$

and therefore $I = a_1 R + a_2 R$. $\square$

Next we prove a characterization in terms of quotient rings. We first need two elementary facts.

Lemma 10. *Let $(A, \mathfrak{m})$ be a commutative Artinian local ring. If every ideal of A is principal, then A is a QF-ring. Conversely, if A is a commutative local QF-ring, then its socle $\text{Soc}(A) = \text{Ann}_A(\mathfrak{m})$ is a simple $A/\mathfrak{m}$ -vector space.*

Proof. Suppose first that every ideal is principal. If $\mathfrak{m} = 0$, then A is a field. Otherwise write $\mathfrak{m} = (a)$ and choose $n \geq 1$ with $\mathfrak{m}^n = 0$ and $\mathfrak{m}^{n-1} \neq 0$. Every ideal is of the form (a^i) , and

$$\text{Ann}_A((a^i)) = (a^{n-i}).$$

Hence every ideal satisfies the double-annihilator property. For a commutative Artinian ring this is equivalent to being quasi-Frobenius. The converse statement, that a local QF-ring has simple socle, is the standard local characterization of commutative QF-rings. $\square$

Lemma 11. *Let R be a commutative ring and let $\mathfrak{m}$ be a maximal ideal. Then $R/\mathfrak{m}^n$ is a local ring with unique prime ideal $\mathfrak{m}/\mathfrak{m}^n$.*

Proof. Prime ideals of $R/\mathfrak{m}^n$ correspond to prime ideals of R containing $\mathfrak{m}^n$. If $\mathfrak{p} \supseteq \mathfrak{m}^n$ is prime, then $\mathfrak{p} \supseteq \sqrt{\mathfrak{m}^n} = \mathfrak{m}$, hence $\mathfrak{p} = \mathfrak{m}$ by maximality. Thus $\mathfrak{m}/\mathfrak{m}^n$ is the unique prime, and therefore the unique maximal ideal, of the quotient. $\square$

Now we can characterize the quotient rings.

Theorem 3. *Let R be a Noetherian domain that is not a field. Then R is a Dedekind domain if and only if every quotient R/I with $0 \neq I \subsetneq R$ is a QF-ring.*

Proof. $\Rightarrow$: Let $0 \neq I \subsetneq R$. Write

$$I = \prod_{i=1}^t \mathfrak{p}_i^{k_i}.$$

By Corollary 4,

$$R/I \cong \prod_{i=1}^t R/\mathfrak{p}_i^{k_i}.$$

Each factor is local by Lemma 11 and is a principal ideal ring by Lemma 9. It is also Artinian: its ideals form the finite chain

$$R/\mathfrak{p}_i^{k_i} \supset \mathfrak{p}_i/\mathfrak{p}_i^{k_i} \supset \cdots \supset \mathfrak{p}_i^{k_i-1}/\mathfrak{p}_i^{k_i} \supset 0.$$

Thus each factor is QF by Lemma 10, and a finite direct product of QF-rings is QF as well.

$\Leftarrow$: Assume every R/I with $0 \neq I \subsetneq R$ is a QF-ring. First, every nonzero prime ideal $\mathfrak{p}$ of R is maximal. Indeed, choose $0 \neq a \in \mathfrak{p}$. Then $R/(a)$ is QF, hence Artinian, and $\mathfrak{p}/(a)$ is therefore maximal in $R/(a)$. Thus $\mathfrak{p}$ is maximal in R , and $\dim R = 1$.

Let $\mathfrak{m}$ be a maximal ideal and put $A = R/\mathfrak{m}^2$. By assumption, A is a local QF-ring. Its maximal ideal $\mathfrak{n} = \mathfrak{m}/\mathfrak{m}^2$ satisfies $\mathfrak{n}^2 = 0$, so $\mathfrak{n} \subseteq \text{Soc}(A)$. Since R is Noetherian and $\mathfrak{m} \neq 0$, Nakayama's lemma gives $\mathfrak{m} \neq \mathfrak{m}^2$, so $\mathfrak{n} \neq 0$. Lemma 10 therefore implies

$$\mathfrak{n} = \text{Soc}(A)$$

and $\mathfrak{m}/\mathfrak{m}^2$ is one-dimensional over $R/\mathfrak{m}$. After localizing at $\mathfrak{m}$, Nakayama's lemma shows that the maximal ideal $\mathfrak{m}R_{\mathfrak{m}}$ is principal. Hence $R_{\mathfrak{m}}$ is a one-dimensional Noetherian local domain with principal maximal ideal, and therefore a DVR, as required. By Lemma 1, R is a Dedekind domain. $\square$

As in Section 2, we now explore injective modules. We first recall the projective-injective duality relevant here.

Lemma 12. *Let R be a commutative ring. If every ideal of R is projective, then every quotient of an injective R -module is injective. More generally, a ring is left hereditary if and only if every quotient of a left injective module is injective.*

Proof. We prove the direction needed below. Let E be injective and let $N \subseteq E$. By Baer's criterion, it is enough to extend any homomorphism $f : I \rightarrow E/N$, where I is an ideal of R , to a homomorphism $R \rightarrow E/N$. Since I is projective, f lifts through the quotient map $E \rightarrow E/N$ to a homomorphism $\tilde{f} : I \rightarrow E$. Since E is injective, $\tilde{f}$ extends to a homomorphism $R \rightarrow E$. Composing with $E \rightarrow E/N$ extends f . Thus E/N is injective. The converse is the standard dual characterization of hereditary rings; see [Rot09]. $\square$

Theorem 4. *Let R be a Dedekind domain. Then every quotient of an injective R -module is injective.*

Proof. By Theorem 2, R is hereditary; equivalently, every ideal of R is projective. The assertion therefore follows immediately from Lemma 12. $\square$

4. FINITELY GENERATED MODULES OVER DEDEKIND DOMAINS

4.1. Torsion-free modules and projective modules. In this subsection, we discuss finitely generated torsion-free modules and their relation to projective modules. We begin by describing them in terms of ideals.

Lemma 13. *Let M be a finitely generated torsion-free module over a Dedekind domain R . Then M is isomorphic to a finite direct sum of nonzero integral ideals of R .*

Proof. Since M is torsion-free, the natural map $M \rightarrow M \otimes_R K$ is injective, where K is the quotient field of R . After choosing a K -basis and clearing denominators of a finite set of generators, we may regard M as a submodule of R^s for some s . We argue by induction on s . For $s = 1$, M is an ideal of R . Suppose $s > 1$ and let $p : R^s \rightarrow R$ be projection onto the last coordinate. Put $\mathfrak{a} = p(M)$ and $N = \ker(p|_M) \subseteq R^{s-1}$. Then

$$0 \rightarrow N \rightarrow M \xrightarrow{p} \mathfrak{a} \rightarrow 0$$

is exact. If $\mathfrak{a} = 0$, then $M = N$ and the induction hypothesis applies. Otherwise $\mathfrak{a}$ is projective by [Theorem 1](#), so the sequence splits and

$$M \cong N \oplus \mathfrak{a}.$$

By induction, N is a finite direct sum of ideals, and the result follows. $\square$

The following lemma connects ideals with the functors $\otimes_R$ and Hom_R .

Lemma 14. *For two nonzero fractional ideals I_1, I_2 of a Dedekind domain R , we have*

- (i) $I_1 \otimes_R I_2 \cong I_1 I_2$;
- (ii) $\text{Hom}_R(I_1, I_2) \cong I_2 I_1^{-1}$.

Proof. (i) Multiplication induces a natural homomorphism

$$I_1 \otimes_R I_2 \longrightarrow I_1 I_2.$$

After localizing at any maximal ideal, both I_1 and I_2 become principal rank-one modules, and the localized multiplication map is an isomorphism. Hence the original map is an isomorphism.

(ii) As in the proof of [Lemma 5](#), every R -linear map $\varphi : I_1 \rightarrow I_2 \subseteq K$ is multiplication by a unique element $a \in K$. Such multiplication maps I_1 into I_2 exactly when $aI_1 \subseteq I_2$, i.e. exactly when $a \in I_2 I_1^{-1}$. This gives the desired isomorphism. $\square$

Remark 1. The concept above generalizes to an arbitrary commutative ring R . An R -module M is called invertible if there exists an R -module N such that $M \otimes_R N \cong R$. Such an M is finitely generated projective of rank one, and N is unique up to isomorphism. The isomorphism classes of invertible modules form a group under tensor product, called the *Picard group* $\text{Pic}(R)$. Moreover,

$$M^{-1} \cong \text{Hom}_R(M, R).$$

For a Dedekind domain, the Picard group agrees with the ideal class group. For details, see [\[AK13, Section 25.18\]](#).

To describe direct sums of ideals further, we use the following standard identity.

Lemma 15. *Let I_1, I_2 be nonzero ideals of a Dedekind domain R . Then*

$$I_1 \oplus I_2 \cong R \oplus I_1 I_2$$

as R -modules.

Proof. Multiplying I_1 by a suitable nonzero element of the quotient field changes neither its isomorphism class as an R -module nor the isomorphism class of $I_1 I_2$, and by the approximation theorem for Dedekind domains we may arrange that $I_1 + I_2 = R$. Then $I_1 \cap I_2 = I_1 I_2$, and

$$0 \longrightarrow I_1 I_2 \longrightarrow I_1 \oplus I_2 \xrightarrow{(x,y) \mapsto x+y} R \longrightarrow 0$$

is exact, where the first map may be taken as $z \mapsto (z, -z)$. Since R is projective, the sequence splits. Therefore

$$I_1 \oplus I_2 \cong R \oplus I_1 I_2.$$

□

We can now classify finite direct sums of ideals.

Lemma 16. *Let R be a Dedekind domain with quotient field K . Let $I_1, \dots, I_n$ and $J_1, \dots, J_m$ be nonzero ideals. The following are equivalent:*

- (i) $I_1 \oplus \dots \oplus I_n \cong J_1 \oplus \dots \oplus J_m$;
- (ii) $n = m$ and there exists $0 \neq \alpha \in K$ such that

$$I_1 \cdots I_n = (\alpha) J_1 \cdots J_n.$$

Proof. $\Leftarrow$: Repeated application of Lemma 15 gives

$$I_1 \oplus \dots \oplus I_n \cong R^{n-1} \oplus I_1 \cdots I_n$$

and similarly for the J_i . Multiplication by α is an isomorphism between the corresponding final ideals, so (i) follows.

$\Rightarrow$: Tensoring the isomorphism in (i) with K gives $K^n \cong K^m$, hence $n = m$. Represent the isomorphism by a matrix $T = (a_{ji})$, where

$$a_{ji} \in \text{Hom}_R(I_i, J_j) \cong J_j I_i^{-1}$$

by Lemma 14. Taking determinants gives

$$(\det T) I_1 \cdots I_n \subseteq J_1 \cdots J_n.$$

Applying the same argument to T^{-1} gives the reverse inclusion, so equality holds. Thus the product ideals differ by multiplication by a nonzero element of K , as required. □

The following is the usual Steinitz normal form.

Corollary 6. *Let M be a nonzero finitely generated torsion-free module over a Dedekind domain R . Then there is a nonzero ideal I of R such that*

$$M \cong R^r \oplus I$$

for some $r \geq 0$. The integer $r + 1 = \text{rank}_R(M)$ and the ideal class $[I]$ are uniquely determined by M .

Proof. By Lemma 13, write $M \cong I_1 \oplus \dots \oplus I_n$. Repeatedly applying Lemma 15 gives

$$M \cong R^{n-1} \oplus I_1 \cdots I_n.$$

After multiplying the last fractional ideal by a nonzero scalar, we may take it to be integral. The uniqueness follows from [Lemma 16](#). $\square$

Torsion-free and projective modules therefore characterize each other in the finitely generated case.

Theorem 5. *Let M be a finitely generated module over a Dedekind domain R . Then M is torsion-free if and only if M is projective.*

Proof. $\Rightarrow$: By [Lemma 13](#), M is a finite direct sum of ideals of R . Each ideal is projective by [Theorem 1](#), so M is projective.

$\Leftarrow$: A projective module is a direct summand of a free module. Since R is a domain, every free module is torsion-free, and every submodule of a torsion-free module is torsion-free. Hence M is torsion-free. $\square$

4.2. Torsion modules and injective modules. We now turn to torsion modules. We first need a useful elementary lemma.

Lemma 17. *A Dedekind domain R with only finitely many nonzero prime ideals is a PID.*

Proof. Let $\mathfrak{p}_1, \dots, \mathfrak{p}_n$ be all the nonzero prime ideals of R . For each i , choose $y_i \in \mathfrak{p}_i \setminus \mathfrak{p}_i^2$. By the Chinese Remainder Theorem, choose $a_i \in R$ such that

$$a_i \equiv y_i \pmod{\mathfrak{p}_i^2}, \quad a_i \equiv 1 \pmod{\mathfrak{p}_j} \quad (j \neq i).$$

Then the prime factorization of (a_i) contains $\mathfrak{p}_i$ with exponent exactly one and contains none of the other nonzero prime ideals. Hence $(a_i) = \mathfrak{p}_i$. Thus every nonzero prime ideal is principal, and unique factorization of ideals shows that every ideal is principal. Therefore R is a PID. $\square$

We can now prove the structure theorem for finitely generated torsion modules over a Dedekind domain.

Lemma 18. *Let $0 \neq M$ be a finitely generated torsion module over a Dedekind domain R . Then there are nonzero proper ideals*

$$I_1 \supseteq I_2 \supseteq \dots \supseteq I_m$$

such that

$$M \cong \bigoplus_{i=1}^m R/I_i.$$

The ideals I_i are uniquely determined by M . Equivalently, M has a unique decomposition into elementary divisors:

$$M \cong \bigoplus_{\mathfrak{p}} \bigoplus_{j=1}^{t_{\mathfrak{p}}} R/\mathfrak{p}^{e_{\mathfrak{p},j}},$$

where only finitely many nonzero prime ideals $\mathfrak{p}$ occur and all $e_{\mathfrak{p},j} \geq 1$; for each $\mathfrak{p}$, the multiset of exponents is uniquely determined.

Proof. Let

$$\mathfrak{a} = \text{Ann}_R(M).$$

Since M is finitely generated and torsion, $\mathfrak{a} \neq 0$. Write

$$\mathfrak{a} = \mathfrak{p}_1^{k_1} \dots \mathfrak{p}_s^{k_s}$$

and let

$$S = R \setminus \bigcup_{i=1}^s \mathfrak{p}_i.$$

The localization $S^{-1}R$ is a Dedekind domain with only finitely many nonzero prime ideals, hence it is a PID by [Lemma 17](#). Moreover, every $s \in S$ maps to a unit of $R/\mathfrak{a}$, and therefore multiplication by s is an automorphism of M . Thus the natural map $M \rightarrow S^{-1}M$ is an isomorphism.

Apply the structure theorem for finitely generated torsion modules over the PID $S^{-1}R$. This gives invariant factors in $S^{-1}R$; contracting them back to R gives ideals $I_1 \supseteq \cdots \supseteq I_m$ and the displayed invariant-factor decomposition over R . Factoring the I_i into prime powers and applying [Corollary 4](#) gives the elementary-divisor form.

For uniqueness, localize at each nonzero prime $\mathfrak{p}$. The module $M_{\mathfrak{p}}$ is a finitely generated torsion module over the DVR $R_{\mathfrak{p}}$, and the ordinary PID structure theorem uniquely determines the exponents $e_{\mathfrak{p},j}$. Hence all elementary divisors, and consequently the invariant factors I_i , are uniquely determined. $\square$

As in the preceding subsection, we next examine the relationship between torsion modules and injective modules. We first recall Matlis' structure theorem.

Lemma 19. (*Matlis*) *Let R be a commutative Noetherian ring. Then:*

- (i) *every nonzero injective R -module is a direct sum of indecomposable injective modules;*
- (ii) *every indecomposable injective R -module is isomorphic to an injective envelope $E(R/\mathfrak{p})$ for a prime ideal $\mathfrak{p}$ of R ;*
- (iii) *the correspondence*

$$\mathfrak{p} \longmapsto E(R/\mathfrak{p})$$

gives a bijection from $\text{Spec}(R)$ to the isomorphism classes of indecomposable injective R -modules.

This is the classical Matlis decomposition theorem from [\[Mat58\]](#); see also [\[Mat86\]](#) and [\[Rot09\]](#).

We will also use a more detailed description of indecomposable torsion modules.

Lemma 20. *Let R be a Dedekind domain. Then*

$$\{R/\mathfrak{p}^n, E(R/\mathfrak{p}) \mid n \geq 1, 0 \neq \mathfrak{p} \in \text{Spec}(R)\}$$

is a complete set of pairwise non-isomorphic indecomposable torsion R -modules.

The classification of indecomposable torsion modules over Dedekind domains is discussed in [\[Du11\]](#) and the references therein.

We can now state the corresponding injectivity properties precisely.

Theorem 6. *Let R be a Dedekind domain. Then:*

- (i) *every indecomposable torsion R -module is quasi-injective;*
- (ii) *if T is an indecomposable torsion R -module, there is a unique nonzero prime ideal $\mathfrak{p}$ such that its injective envelope satisfies*

$$E(T) \cong E(R/\mathfrak{p});$$

in particular, $E(T)$ is indecomposable;

(iii) every torsion R -module has a torsion injective envelope, which is a direct sum of modules $E(R/\mathfrak{p})$ for nonzero prime ideals $\mathfrak{p}$. In particular, if T is finitely generated and

$$T \cong \bigoplus_{\mathfrak{p}} \bigoplus_{j=1}^{t_{\mathfrak{p}}} R/\mathfrak{p}^{e_{\mathfrak{p},j}},$$

then

$$E(T) \cong \bigoplus_{\mathfrak{p}} E(R/\mathfrak{p})^{\oplus t_{\mathfrak{p}}}.$$

Proof. (i) By Lemma 20, an indecomposable torsion module is either $R/\mathfrak{p}^n$ or $E(R/\mathfrak{p})$. The latter is injective. For the former, $R/\mathfrak{p}^n$ is a QF-ring by Theorem 3, hence it is injective as a module over itself. Every R -submodule and every R -linear map between submodules is automatically $R/\mathfrak{p}^n$ -linear, so $R/\mathfrak{p}^n$ is quasi-injective as an R -module.

(ii) If $T = E(R/\mathfrak{p})$, the claim is immediate. If $T = R/\mathfrak{p}^n$, then its socle $\mathfrak{p}^{n-1}/\mathfrak{p}^n \cong R/\mathfrak{p}$ is essential in T . Hence the injective envelope of T is the injective envelope of $R/\mathfrak{p}$, namely $E(R/\mathfrak{p})$. Uniqueness follows from Lemma 19.

(iii) Every module has an injective envelope. By Lemma 19, $E(T)$ decomposes as a direct sum of modules $E(R/\mathfrak{q})$. Since T is essential in $E(T)$ and torsion, no summand corresponding to the zero prime can occur: such a summand is torsion-free and would meet T trivially. Hence only nonzero primes occur, and $E(T)$ is torsion. For finitely generated T , apply part (ii) to each elementary-divisor summand in Lemma 18. A finite direct sum of injective modules is injective, and the direct sum of the corresponding essential extensions is again essential, yielding the displayed formula. $\square$

4.3. Structure theorem of finitely generated modules. We conclude by combining the torsion-free and torsion parts.

Lemma 21. *Let R be a Dedekind domain, let M be a finitely generated R -module, and let $T(M)$ be its torsion submodule. Then there exists a torsion-free submodule $G \subseteq M$ such that*

$$M = T(M) \oplus G.$$

Moreover, $G \cong M/T(M)$, so G is unique up to isomorphism.

Proof. There is a short exact sequence

$$0 \longrightarrow T(M) \longrightarrow M \longrightarrow M/T(M) \longrightarrow 0.$$

The quotient $M/T(M)$ is finitely generated and torsion-free, hence projective by Theorem 5. Therefore the sequence splits, giving

$$M \cong T(M) \oplus G$$

with $G \cong M/T(M)$ torsion-free. The isomorphism class of G is therefore uniquely determined. $\square$

Finally, we obtain the structure theorem.

Theorem 7. *Let R be a Dedekind domain and let M be a finitely generated R -module. Put $r = \text{rank}_R(M)$. If $r \geq 1$, there is a nonzero ideal I of R such that*

$$M \cong R^{r-1} \oplus I \oplus \bigoplus_{\mathfrak{p}} \bigoplus_{j=1}^{t_{\mathfrak{p}}} R/\mathfrak{p}^{e_{\mathfrak{p},j}},$$

where only finitely many nonzero prime ideals $\mathfrak{p}$ occur and $e_{\mathfrak{p},j} \geq 1$. If $r = 0$, the first two summands are omitted. The rank r , the ideal class $[I]$ when $r \geq 1$, and, for every $\mathfrak{p}$, the multiset of exponents $\{e_{\mathfrak{p},j}\}_j$ are uniquely determined by M .

Proof. By Lemma 21,

$$M \cong T(M) \oplus M/T(M).$$

Apply Corollary 6 to the torsion-free summand and Lemma 18 to the torsion summand. Their respective uniqueness statements give the uniqueness asserted above.

□

REFERENCES

- [AF92] Frank W. Anderson and Kent R. Fuller. *Rings and Categories of Modules*. Springer, 1992.
- [AK13] Allen Altman and Steven Kleiman. *A Term of Commutative Algebra*. Worldwide Center of Mathematics, LLC, 2013.
- [AM94] M. F. Atiyah and I. G. Macdonald. *Introduction to Commutative Algebra*. Westview Press, 1994.
- [Aus74] M. Auslander. *Representation Theory of Artin Algebras*. Communications in Algebra, **1** (1974), 269–310.
- [Bas62] H. Bass. *Torsion Free and Projective Modules*. Transactions of the American Mathematical Society, **102**(2) (1962), 319–327.
- [BMO10] D. Bennis, N. Mahdou, and K. Ouarghi. *Rings over which all modules are strongly Gorenstein projective*. Rocky Mountain Journal of Mathematics, **40**(3) (2010), 749–759.
- [Du11] Jiachun Du. *Indecomposable Torsion Modules over Dedekind Domains*. Journal of Mathematical Research and Exposition, **31** (2011). DOI: 10.3770/j.issn:1000-341X.2011.01.008.
- [Fen86] Feng Keqin. *Fundamentals of Commutative Algebra*. Higher Education Press, 1986.
- [Hu12] Hu Kui. *Gorenstein Projective Modules over Domains*. Master’s thesis, Sichuan Normal University, 2012.
- [Kap69] I. Kaplansky. *Infinite Abelian Groups*. University of Michigan Press, Ann Arbor, 1969.
- [KOS07] Leonid A. Kurdachenko, Javier Otal, and Igor Ya. Subbotin. *Artinian Modules over Group Rings*. Birkhäuser, 2007.
- [Mat58] Eben Matlis. *Injective Modules over Noetherian Rings*. Pacific Journal of Mathematics, **8** (1958), 511–528. DOI: 10.2140/pjm.1958.8.511.
- [Mat59] Eben Matlis. *Injective Modules over Prüfer Rings*. Nagoya Mathematical Journal, **15** (1959), 57–69. DOI: 10.1017/S002776300000667X.
- [Mat60] Eben Matlis. *Divisible Modules*. Proceedings of the American Mathematical Society, **11** (1960), 385–391. DOI: 10.1090/S0002-9939-1960-0116044-8.
- [Mat86] H. Matsumura. *Commutative Ring Theory*. Cambridge University Press, Cambridge, 1986.
- [Rot09] Joseph J. Rotman. *An Introduction to Homological Algebra*. Springer, 2009.
- [Stacks] The Stacks Project Authors. *The Stacks Project*. See Lemma 10.120.17 (Tag 034X), Lemma 10.78.2 (Tag 00NX), and Lemma 10.78.5 (Tag 00NZ). <https://stacks.math.columbia.edu/>.